Faculty of Technology & Engineering
Bachelor of Technology Programme
Computer Engineering
(B.Tech. CE)

ACADEMIC REGULATIONS
&
SYLLABUS

(Choice Based Credit System)

Academic Year: 2024-2025

Chandubhai S. Patel Institute of Technology

Vision

To become a leading institute for creation and dissemination of knowledge in the frontiers of Technology.

Mission

To prepare world-class technocrats and researchers and facilitate enhanced deployment of technology for betterment of lives.

Department of Computer Engineering

Vision

To emerge as an eminent research driven entity in Education in the field of computer engineering to meet the needs of the fast paced modern society.

Mission

To develop competent and responsible Computer Professionals for amelioration of the society at large.

THE PROGRAMME EDUCATIONAL OBJECTIVES (PEOs)

PEO1	To prepare the student(s) for successful career as an engineer, a corporate or a government professional, a scientist, ac academician, a technocrat, an administrator and an entrepreneur.
PEO2	To make students demonstrate their abilities to adapt to a rapidly changing environment by having learning approach and apply new skills and new technologies to solve the problems.
PEO3	To create an ambience where the students are cared for in every aspect and motivated to become excellent working professionals who will continue to cherish their association with the organization as a whole, staff and colleagues.
PEO4	To provide continued professional development and lifelong learning throughout their career to inculcate strong teamwork and by installing lacking skills for the benefit of the society.
PEO5	To prepare the students to apply their technical skill(s) to analyse and design appropriate solution(s) with social consciousness and ethical values.

PROGRAM ARTICULATION MATRIX

(-Not Matching,1-Low,2,3-High)

Mission Statement	PEO1	PEO2	PEO3	PEO4	PEO5
To provide world-class educational activities	3	3	3	3	3
To deploy latest technologies	-	2	3	3	-
To foster research activities in consultation with academia and industry	3	2	2	3	3

PROGRAM OUTCOMES (POs)

At the end of the program, the student will be able to:

PO1	**Engineering knowledge:** Apply knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.
PO2	**Problem analysis:** Identify, formulate, review research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences.
PO3	**Design/development of solutions:** Design solutions for complex engineering problems and design system components or processes that meet specified needs with appropriate consideration for public health and safety, and the cultural, societal, and environmental considerations.
PO4	**Conduct investigations of complex problems:** Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.
PO5	**Modern tool usage:** Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.
PO6	**The engineer and society:** Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.
PO7	**Environment and sustainability:** Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.
PO8	**Ethics:** Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.
PO9	**Individual and team work**: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.
PO10	**Communication:** Communicate effectively on complex engineering activities with the engineering community and with society at large, such as being able to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.
PO11	**Project management and finance:** Demonstrate knowledge and understanding of the engineering and management principles and apply these to one’s own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.
PO12	**Life-long learning:** Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES (PSOs)

At the end of the program, the student will be able to:

PSO1	Apply good analytical, design and implementation skills required to formulate and solving computational problems.
PSO2	Excellent adaptability to function in multi-disciplinary work environment, good interpersonal skills in appreciation of professional ethics and societal responsibilities.

FACULTY OF TECHNOLOGY AND ENGINEERING
ACADEMIC REGULATIONS
Bachelor of Technology Programmes
Choice Based Credit System

To ensure uniform system of education, duration of undergraduate and post graduate programmes, eligibility criteria for and mode of admission, credit load requirement and its distribution between course and system of examination and other related aspects, following academic rules and regulations are recommended.

1. *System of Education*

Choice based Credit System with Semester pattern of education shall be followed across The Charotar University of Science and Technology (CHARUSAT) both at Undergraduate and Master's levels. Each semester will be at least 90 working day duration. Every enrolled student will be required to take a course works in the chosen subject of specialization and also complete a project/dissertation if any. Apart from the Programme Core courses, provision for choosing University level electives and Programme/Institutional level electives are available under the Choice based credit system.

2. *Duration of Programme*

(i)	Undergraduate programme	(B. Tech.)
	Minimum	8 semesters (4 academic years)
	Maximum	16 semesters (8 academic years)

3. *Eligibility for admissions*

As enacted by Govt. of Gujarat from time to time.

4. *Mode of admissions*

As enacted by Govt. of Gujarat from time to time.

5. *Programme structure and Credits*

As per annexure – 1 attached

6. *Attendance*

6.1 All activities prescribed under these regulations and listed by the course faculty members in their respective course outlines are compulsory for all students pursuing the courses. No exemption will be given to any student from attendance except on account of serious personal illness or accident or family calamity that may genuinely prevent a student from attending a particular session or a few sessions. However, such unexpected absence from classes and other activities will be required to be condoned by the Dean/Principal.

6.2 Student attendance in a course should be Minimum 80%.

7 *Course Evaluation*

7.1 The performance of every student in each course will be evaluated as follows:

7.1.1 Internal evaluation by the course faculty member(s) based on continuous assessment, for 30% of the marks for the course; and

7.1.2 Final examination by the University through written paper or practical test or oral test or presentation by the student or a combination of any two or more of these, for 70% of the marks for the course.

7.2 *University Examination*

7.2.1 The final examination by the University for 70% of the evaluation for the course will be through written paper and 100% for practical test or oral test or presentation by the student or a combination of any two or more of these.

7.2.2 **In order to earn the credit in a course a student has to obtain grade other than FF.**

7.3 Performance at Internal & University Examination

7.3.1 Minimum performance with respect to internal marks as well as university examination will be an important consideration for passing a course. Details of minimum percentage of marks to be obtained in the examinations (internal/external) are as follows

Minimum marks in University Exam per subject	Minimum marks Overall per subject
40%	45%

7.3.2 A student failing to score 45% of the final examination will get a FF grade.

7.3.3 If a candidate obtains minimum required marks per subject but fails to obtain minimum required overall marks, he/she has to repeat the university examination till the minimum required overall marks are obtained.

8 Grading

8.1 The total of the internal evaluation marks and final University examination marks in each course will be converted to a letter grade on a ten-point scale as per the following scheme:

Table: Grading Scheme (UG)

Range of Marks (%)	**≥80**	**≥73 <80**	**≥66 <73**	**≥60 <66**	**≥55 <60**	**≥50 <55**	**≥45 <50**	**<45**
Letter Grade	**AA**	**AB**	**BB**	**BC**	**CC**	**CD**	**DD**	**FF**
Grade Point	**10**	**9**	**8**	**7**	**6**	**5**	**4**	**0**

8.2 The student's performance in any semester will be assessed by the Semester Grade Point Average (SGPA). Similarly, his performance at the end of two or more consecutive semesters will be denoted by the Cumulative Grade Point Average (CGPA). The SGPA and CGPA are calculated as follows:

(i) SGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses in the semester

(ii) CGPA = $\sum C_i G_i / \sum C_i$ where C_i is the number of credits of course i
G_i is the Grade Point for the course i
and i = 1 to n, n = number of courses of all semesters up to which CGPA is computed.

(iii) No student will be allowed to move further if CGPA is less than 3 at the end of every academic year.

(iv) A student will not be allowed to move to third year if he/she has not cleared all the courses of first year.

(v) A student will not be allowed to move to fourth year if he/she has not cleared all the courses of second year.

9. Awards of Degree

9.1 Every student of the programme who fulfils the following criteria will be eligible for the award of the degree:

9.1.1 He should have earned at least minimum required credits as prescribed in course structure; and

9.1.2 He should have cleared all internal and external evaluation components in every course; and

9.1.3 He should have secured a minimum CGPA of 4.5 at the end of the programme;

9.1.4 In addition to above, the student has to complete the required formalities as per the regulatory bodies, if any.

9.2 The student who fails to satisfy minimum requirement of CGPA at the end of program will be allowed to improve the grades so as to secure a minimum CGPA for award of degree. Only latest grade will be considered.

10. Award of Class

The class awarded to a student in the programme is decided by the final CGPA as per the following scheme:

Distinction:	CGPA $\geq$ 7.5
First class:	CGPA$\geq$ 6.0
Second Class:	CGPA$\geq$ 5.0

11. Transcript

The transcript issued to the student at the time of leaving the University will contain a consolidated record of all the courses taken, credits earned, grades obtained, SGPA, CGPA, class obtained, etc.

Choice Based Credit System

With the aim of incorporating the various guidelines initiated by the University Grants Commission (UGC) to bring equality, efficiency and excellence in the Higher Education System, Choice Based Credit System (CBCS) has been adopted. CBCS offers wide range of choices to students in all semesters to choose the courses based on their aptitude and career objectives. It accelerates the teaching-learning process and provides flexibility to students to opt for the courses of their choice and / or undergo additional courses to strengthen their Knowledge, Skills and Attitude.

1. CBCS – Conceptual Definitions / Key Terms (Terminologies)

1.1. Core Courses

1.1.1 University Core (UC)

University Core Courses are those courses which all students of the University of a Particular Level (PG/UG) will study irrespective of their Programme/specialisation.

1.1.2 Programme Core (PC)

A ‘Core Course’ is a course which acts as a fundamental or conceptual base for Chosen Specialisation of Engineering. It is mandatory for all students of a particular Programme and will not have any other choice for the same.

1.2 Elective Course (EC)

An ‘Elective Course’ is a course in which options / choices for course will be offered. It can either be for a Functional Course / Area or Streams of Specialization / Concentration which is / are offered or decided or declared by the University/Institute/Department (as the case may be) from time to time.

1.2.1 Institute Elective Course (IE)

Institute Courses are those courses which any students of the University/Institute of a Particular Level (PG/UG) will choose as offered or decided by the University/Institute from time-to-time irrespective of their Programme /Specialisation

1.2.2 Programme Elective Course (PE):

A ‘Programme Elective Course’ is a course for the specific programme in which students will opt for specific course(s) from the given set of functional course/ Area or Streams of Specialization options as offered or decided by the department from time-to-time

1.2.3 Cluster Elective Course (CE):

An ‘Elective Course’ is a course which students can choose from the given set of functional course/ Area or Streams of Specialization options (eg.

Common Courses to EC/CE/IT/EE) as offered or decided by the Institute from time-to-time.

1.3 Non-Credit Course (NC) - AUDIT Course

A ‘Non-Credit Course’ is a course where students will receive Participation or Course Completion certificate. This will not be reflected in Student’s Grade Sheet. Attendance and Course Assessment is compulsory for Non-Credit Courses

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY												
TEACHING & EXAMINATION SCHEME FOR B TECH PROGRAMME IN CE ENGINEERING												
CHOICE BASED CREDIT SYSTEM												
	Course Code	Course Title	Teaching Scheme				Examination Scheme					
			Contact Hours			Credit	Theory		Practical		Total
			Theory	Practical	Total		Internal	External	Internal	External	
Sem 7	CE442	Design of Language Processors	4	2	6	5	30	70	25	25	150
	CE443	Cloud Computing	3	2	5	4	30	70	25	25	150
	CE449	Big Data Analytics	3	2	5	4	30	70	25	25	150
	CEXXX	Programme Elective-III	3	2	5	4	30	70	25	25	150
	CE450	Software Group Project-V	0	2	4	2	0	0	50	50	100
	CE446	Summer Internship-II	0	0	0	3	0	0	75	75	150
			13	**10**	**25**	**22**	**120**	**280**	**225**	**225**	**850**
Sem 8	CE451	Software Project Major	0	36	36	18	0	0	250	350	600

Note:

- **University Elective (UE):-** University Electives are offered in common slots and offered by various departments. Students of any programme can select these electives. Subjects like Research Methodology, Occupational Health & Safety, Engineering Economics, Professional Ethics, and Project Management, Disaster Management, Risk Management etc. can be included.
- **Cluster Elective (CT):-** Institutional Electives means common electives among a cluster of programmes (eg. CE/IT/EC/EE etc.). If Institutional Electives are not applicable, it will be Programme electives
- **Programme Elective (PE):-**
- **Institute Elective (IE):-**
- Provision for Auditing a course will be available
- Audit courses may be offered and decided based on need of the institute/program(s)

List of Programme Electives		
ELECTIVES	**Code**	**Programme Elective - III**
	CE477	Internet of Things
	CE478	Natural Language Processing
	CE479	Ethical Hacking

B. Tech. (Computer Engineering) Programme

SYLLABI

(Semester – 7)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE442: DESIGN OF LANGUAGE PROCESSOR

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	4	2	-	6	**5**
Marks	100	50	-	150	

Pre-requisite courses:

- Digital Electronics
- Operating System
- Theory of Computation

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Overview of Language Processors & Translators	10
2.	Introduction To Compilers	20
3.	Symbol-Table Management	05
4.	Static & Dynamic Memory Allocation & Memory management	06
5.	Semantic Analysis & Intermediate Code Generation	05
6.	Code Optimization	07
7.	Code Generation	07
	Total hours (Theory) :	60
	Total hours (Lab) :	30
	Total hours :	90

Detailed Syllabus:

1.	**Overview of Language Processors & Translators**	**10 Hours**	**20%**
	Language processing activities, fundamental of language processing, Operating System, Interpreter vs compiler Pre-processor & Macro Processors – Subroutine, Macro definition and call, Macro expression, nested macro call, Advanced macro facilities, design of macro pre-processor Loaders - Compile & go, Absolute, Bootstrap, Relocating, Linking loader Linkers - Relocation of Linking Concept, Design of Linker,		

	Linker for MS DOS, Linking for overlays, Linkage editor Assemblers - Elements of Assembly Language Programming, Assembly Scheme, Single pass Assembler, Two pass assembler, Data structure of Assemblers		
2.	**Introduction To Compilers**	**20 Hours**	**35%**
	Pass and Phases of Compiler, grouping of phases, Compiler Contraction tools Lexical analyser - Roles of lexical analyser, input buffering, tokens, Regular Expression, finite Automata Syntax analyser - Context free grammar, Ambiguous grammar, Top-down parsing, Bottom-up parsing, LEX, YACC		
3.	**Symbol-Table Management**	**05 Hours**	**05%**
	Data structures to implement symbol table, Symbol Attributes, Symbol-Table entries, Local Symbol Table management, Global Symbol Table Structure, Storage bindings and Symbolic Registers		
4.	**Static & Dynamic Memory Allocation & Memory management**	**06 Hours**	**09%**
	Data descriptors - Static and Dynamic storage allocation – Storage allocation and access in block structured programming languages – Array allocation and access- Compilation of expressions – Handling operator priorities – Intermediate code forms for expressions – code generator., Register Usage, The run-time stack, Parameter passing disciplines, Code sharing and Position-Independent code		
5.	**Semantic Analysis & Intermediate Code Generation**	**05 Hours**	**09%**
	Intermediate Languages, Declarations, Assignment Statements, Boolean Expressions, Case Statements, Back patching, Procedure Calls		
6.	**Code Optimization**	**07 Hours**	**11%**
	The Principal Sources of Optimization, Optimization of Basic Blocks, Loops in Flow Graphs, Introduction to Global Data-Flow Analysis, Iterative Solution of Data-Flow Equations,		

	Code-Improving Transformations, Dealing with Aliases, Data-Flow Analysis of Structured Flow Graphs, Efficient Data-Flow Algorithms, A Tool for Data-Flow Analysis, Estimation of Types, Symbolic Debugging of Optimized Code		
7.	**Code Generation**	**07 Hours**	**11%**
	Issues in the Design of a Code Generator, The Target Machine, Run-Time Storage Management, Basic Blocks and Flow Graphs, Next-Use Information, A Simple Code Generator, Register Allocation and Assignment, The DAG Representation of Basic Blocks, Peephole Optimization, Generating Code from DAGs, Dynamic Programming Code-Generation Algorithm, Code-Generator Generators		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand design and processing of different language processor, loaders and linkers
CO2	Design top-down and bottom-up parsers
CO3	Identify different memory management schemes of language processors
CO4	Develop semantic analysis scheme to generate intermediate code
CO5	Apply different code optimization techniques
CO6	Develop algorithms to generate code for a target machine

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	-	-	-	-	-	-	-	-	-	-	-	-	-
CO2	1	-	-	3	-	-	-	-	-	-	-	-	-	-
CO3	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO4	2	2	2	-	2	-	-	-	-	-	-	-	-	-
CO5	2	-	-	-	-	-	-	-	-	-	-	-	-	-
CO6	2	-	-	1	-	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- **Text book:**
    1. Alfred Aho, Ravi Sethi, Jeffrey D Ullman, "Compilers Principles, Techniques and- Tools", Pearson Education Asia.
    2. M. Dhamdhere, "System Programming and Operating Systems", Tata McGraw-Hill.
    3. Steven S. Muchnick. Advanced Compiler Design and Implementation
- **Reference book:**
    1. Allen I. Holub "Compiler Design in C", Prentice Hall of India.
    2. C. N. Fischer and R. J. LeBlanc, "Crafting a compiler with C", Benjamin Cummings.
    3. J.P. Bennet, "Introduction to Compiler Techniques", Second Edition, Tata McGraw-Hill
    4. HenkAlblas and Albert Nymeyer, "Practice and Principles of Compiler Building with C", PHI.
    5. Kenneth C. Louden, "Compiler Construction: Principles and Practice", Thompson Learning.
    6. Compiler Construction by Kenneth. C. Louden, Vikas Pub
- **Web material:**
    1. http://compilers.iecc.com/crenshaw
    2. http://www.compilerconnection.com
    3. http://dinosaur.compilertools.net
    4. http://pltplp.net/lex-yacc
- **Software:**
    1. LEX
    2. YACC

CE443: CLOUD COMPUTING

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	**4**
Marks	100	50	-	150	

Pre-requisite courses:

- Operating System
- Networking

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Cluster Computing, Grid Computing Systems and Resource Management at Glance	08
2.	Fundamental of Virtualization	06
3.	Fundamental Concepts and Models	06
4.	Cloud-Enabling Technology	05
5.	Fundamental Cloud Architectures	07
6.	Advanced Cloud Architectures	08
7.	Implementation of Cloud	05
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	**Cluster Computing, Grid Computing Systems and Resource Management at Glance**	**08 Hours**	**18%**
	Introduction, Eras of Computing, Scalable Parallel Computer Architectures, Towards Low Cost Parallel Computing and Motivations , A Cluster Computer and its Architecture, Clusters Classification, Commodity Components for Clusters, Grid Architecture and Service Modelling, Grid Projects and Grid Systems Built, Grid Resource Management and Brokering, Software and Middleware for Grid Computing,		

	Grid Application Trends		
2.	**Fundamental of Virtualization**	**06 Hours**	**12%**
	Type of Virtualization, Virtualization Technologies, Virtualizes your Environment, Managing Virtualization Environment, Storage Virtualization, Dockers		
3.	**Fundamental Concepts and Models**	**06 Hours**	**15%**
	Roles and Boundaries, Cloud Characteristics, Cloud Delivery Models, Cloud Deployment Models		
4.	**Cloud-Enabling Technology**	**05 Hours**	**14%**
	Broadband Networks and Internet Architecture, Data centre Technology, Virtualization Technology, Web Technology, Multitenant Technology, Service Technology		
5.	**Fundamental Cloud Architectures**	**07 Hours**	**16%**
	Workload Distribution Architecture, Resource Pooling Architecture ,Dynamic Scalability Architecture, Elastic Resource Capacity Architecture, Service Load Balancing Architecture, Cloud Bursting Architecture, Elastic Disk Provisioning Architecture, Redundant Storage Architecture		
6.	**Advanced Cloud Architectures**	**08 Hours**	**17%**
	Hypervisor Clustering Architecture ,Load Balanced Virtual Server Instances Architecture, Non-Disruptive Service Relocation Architecture, Zero Downtime Architecture ,Cloud Balancing Architecture ,Resource Reservation Architecture, Dynamic Failure Detection and Recovery Architecture		
7.	**Implementation of Cloud**	**05 Hours**	**08%**
	Study of Cloud computing Systems like Amazon EC2 and S3, Google App Engine, and Microsoft Azure, Build Private/Hybrid Cloud using open source tools, Deployment of Web Services from Inside and Outside a Cloud Architecture. MapReduce and its extensions to Cloud Computing, HDFS, and GFS		

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Assess and examine advantages and disadvantages of virtualization technology.
CO2	Compose services in a distributed computing environment to achieve tasks relevant to a knowledge-based business or public service
CO3	Evaluate a set of business requirements to determine suitability for a cloud computing delivery model.
CO4	Explore the various cloud computing architectures and paradigms.
CO5	Deployment of cloud and identify security implications in cloud computing.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	2	2	2	1	0	1	-	-	-	1	2	-
CO2	3	3	3	1	3	2	1	-	1	-	1	3	2	1
CO3	3	3	2	3	3	1	2	-	2	1	-	3	2	1
CO4	3	2	1	2	2	-	-	2	1	-	-	2	2	2
CO5	3	2	3	1	3	2	1	1	2	1	-	3	1	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- ❖ **Text Books:**
  1. Thomas Erl, Zaigham Mahmood,and Ricardo Puttini, “Cloud Computing Concepts, Technology & Architecture”, Prentice Hall
  2. Kai Hwang, Geoffrey C.,” Distributed and Cloud Computing”, Morgan Kaufmann is an imprint of Elsevier
  3. Navin Sabharwal, Ravi Shankar “Apache CloudStack Cloud Computing” PACKT Publishing

- ❖ **Reference Books**:

1. Ravi Shankar, Navin Sabharwa “Cloud Computing First Steps: Cloud Computing for Beginners” Create Space Independent Publishing Platform
2. Rajkumar Buyya, James Broberg, Andrzej Goscinski “Cloud Computing: Principles and Paradigms” Wiley
3. Judith Hurwitz, Robin Bloor “Cloud Computing For Dummies” , for Dummies

- **Web material:**

1. http://www.console.cloud.google.com
2. http://www.qwicklabscom
3. http://codelabs.developers.google.com
4. http://www.docker.com

- **Software/Platform:**

1. NetBeans
2. Eclipse
3. .NET
4. GCP
5. Amazone
6. Microsoft

CE449: BIG DATA ANALYTICS

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	3	2	-	5	**4**
Marks	100	50	-	150	

Pre-requisite courses:

- Linux Operating System
- Database Management System

Outline of the Course:

Sr. No.	Title of the unit	Minimum number of hours
1.	Big Data and Analytics	02
2.	Introduction to Hadoop and Hadoop Architecture	07
3.	HDFS, HIVE AND HBASE	08
4.	Apache Spark	10
5.	Spark SQL and Spark Streaming	10
6.	Graph Analytics and Data Visualization	08
	Total hours (Theory) :	45
	Total hours (Lab) :	30
	Total hours :	75

Detailed Syllabus:

1.	**Big Data and Analytics**	**02 Hours**	**10%**
	Introduction to Big Data, Big Data Characteristics, Types of Big Data, Traditional Versus Big Data Approach, Technologies Available for Big Data, Infrastructure for Big Data, Use of Data Analytics, Big Data Challenges.		
2.	**Introduction to Hadoop and Hadoop Architecture**	**07 Hours**	**15%**
	Big Data – Apache Hadoop & Hadoop EcoSystem, Moving Data in and out of Hadoop – Understanding inputs and outputs in Hadoop, Data Serialization		

3.	**HDFS, HIVE AND HBASE**	**08 Hours**	**20%**
	HDFS-Overview, Installation and Shell, Hive Architecture and Installation, Comparison with Traditional Database, HiveQL Querying Data, Sorting And Aggregating, Map Reduce Scripts, Joins & Sub queries, HBase concepts, Advanced Usage, Schema Design, Advance Indexing		
4.	**Apache Spark**	**10 Hours**	**20%**
	Introduction to Data Analysis with Spark, Downloading Spark and Getting Started, Apache Spark components and API stack, Application and Spark Session, Introduction to RDD, RDD and Data Frames.		
5.	**Spark SQL and Spark Streaming**	**10 Hours**	**20%**
	Big Data and Spark SQL, Spark-Managed Tables, Reading Tables into Data Frames, Aggregations, Joins, Creating Views, Spark Streaming and Challenges of Stream Processing, Spark's Streaming APIs, Spark streaming case study.		
6.	**Graph Analytics and Data Visualization**	**08 Hours**	**15%**
	Apache Spark GraphX: Property Graph, Graph Operator, SubGraph, Triplet, Neo4j: Modeling data with Neo4j, Cypher Query Language: General clauses, Read and Write clauses. Big Data Visualization with Power BI, Apache Super-Set		

At the end of the course, the students will be able to

CO1	Understand the key issues in big data management and its associated applications in intelligent business and scientific computing
CO2	Acquire fundamental enabling techniques and scalable algorithms like Hadoop, Map Reduce, Hive and Spark in big data analytics.
CO3	Evaluate and apply appropriate principles, techniques and theories to large-scale data science problems using various databases with analytics and visualizations.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	2	1	-	-	-	-	-	-	-	-	-	1	1
CO2	1	2	3	1	3	-	-	-	-	-	-	-	2	-
CO3	-	1	3	3	3	-	-	-	-	-	-	-	2	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material:

- ❖ **Text book:**
  1. Bart Baesens , Analytics in a Big Data World: The Essential Guide to Data Science and its Applications, ,Wiley, 2014
  2. Jules S. Damji, Learning SparkLightning-Fast Data Analytics, O'Reilly Media Inc, 2020.
  3. Spark: The Definitive Guide by Bill Chambers and Matei Zaharia, O'Reilly Media Inc. 2018.

- ❖ **Reference book:**
  1. Xyz Dirk Deroos et al., Hadoop for Dummies, Dreamtech Press, 2014.
  2. Chuck Lam, Hadoop in Action, December, 2010.
  3. Leskovec, Rajaraman, Ullman, Mining of Massive Datasets, Cambridge University Press.
  4. I.H. Witten and E. Frank, Data Mining: Practical Machine learning tools and techniques.

- ❖ **Web material:**
  1. https://cognitiveclass.ai/
  2. https://codelabs.developers.google.com/

- ❖ **Software & Platform:**
  1. R & SPSS
  2. Hadoop, HBase, Hive, Pig, Spark
  3. Casandra, Neo4j, NoSQL

CE477: INTERNET OF THINGS (PE-III)

Description:

This course OCCE3003–Introduction to Internet of Things is offered from SWAYAM as noc23_cs83 – Introduction to Internet of Things

URL : https://onlinecourses.nptel.ac.in/noc23_cs83/preview

Credits and Hours:

Teaching Scheme	Week	Theory	Practical	Total
Credit	12	3	1	4
Marks		100	50	150

***** Practical component is offered by CHARUSAT

About this course:

The Internet of Things (IoT) is presently a hot technology worldwide. Government, academia, and industry are involved in different aspects of research, implementation, and business with IoT. IoT cuts across different application domain verticals ranging from civilian to defence sectors. These domains include agriculture, space, healthcare, manufacturing, construction, water, and mining, which are presently transitioning their legacy infrastructure to support IoT. Today it is possible to envision pervasive connectivity, storage, and computation, which, in turn, gives rise to building different IoT solutions. IoT-based applications such as innovative shopping systems, infrastructure management in both urban and rural areas, remote health monitoring and emergency notification systems, and transportation systems, are gradually relying on IoT-based systems. Therefore, it is very important to learn the fundamentals of this emerging technology.

Course Layout:

Week 1 : Introduction to IoT: Part I, Part II, Sensing, Actuation, Basics of Networking: Part I

Week 2 : Basics of Networking: Part II, Part III, Part IV, Communication Protocols: Part I, Part II

Week 3 : Communication Protocols: Part III, Part IV, Part V, Sensor Networks: Part I, Part II

Week 4 : Sensor Networks: Part III, Part IV, Part V, Part VI, Machine-to-Machine Communications

Week 5 : Interoperability in IoT, Introduction to Arduino Programming: Part I, Part II, Integration of Sensors and Actuators with Arduino: Part I, Part II

Week 6 : Introduction to Python programming, Introduction to Raspberry Pi, Implementation of IoT with Raspberry Pi

Week 7 : Implementation of IoT with Raspberry Pi (contd), Introduction to SDN, SDN for IoT

Week 8 : SDN for IoT (contd), Data Handling and Analytics, Cloud Computing

Week 9 : Cloud Computing(contd), Sensor-Cloud

Week 10 : Fog Computing, Smart Cities and Smart Homes

Week 11 : Connected Vehicles, Smart Grid, Industrial IoT

Week 12 : Industrial IoT (contd), Case Study: Agriculture, Healthcare, Activity Monitoring

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Interpret the design issues of IoT & M2M
CO2	Understand the protocol stack specific to sensor network and IoT applications
CO3	Demonstrate a critical understanding of key technologies in data transmission and data processing for IoT.
CO4	Evaluation of privacy management, challenges in designing and securing an IoT system.
CO5	Illustrate the application of IoT in industrial automation and identify real world design Constraints.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	1	2	2	-	-	-	-	-	-	-	-	-	-	-
CO2	3	2	3	2	1	-	-	-	-	-	-	-	3	-
CO3	1	2	3	-	1	-	-	-	-	-	-	-	2	-
CO4	-	1	2	1	1	-	-	-	-	-	-	-	2	-
CO5	2	1	3	-	1	-	-	-	-	-	-	-	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Books and References:

- S. Misra, A. Mukherjee, and A. Roy, 2020. *Introduction to IoT*. Cambridge University Press.
- S. Misra, C. Roy, and A. Mukherjee, 2020. *Introduction to Industrial Internet of Things and Industry 4.0.* CRC Press.
- Internet of Things: Architectures, Protocols and Standards 1st Edition , Simone Cirani, Gianluigi Ferrari, Marco Picone, and Luca Veltri.
- Internet of Things: principles and paradigms, Buyya, Rajkumar and Amir Vahid Dasterdji (eds.), Morgan Kaufmann, 2016.
- From Machine-to-Machine to the Internet of Things: introduction to a new age of intelligence, Holler, Jan et al, Academic Press, 2014.
- The Internet of Things Enabling Technologies, Platforms, and Use Cases, Pethuru Raj Anupama C. Raman,2017
- Building Internet of Things with the Arduino, Doukas, Charalampos, Create Space Independent Publishing Platform, 2012.
- Francis daCosta, "Rethinking the Internet of Things: A Scalable Approach to Connecting Everything", 1st Edition, Apress Publications, 2013.
- Research Papers

- **Web material:**
    1. http://web.mit.edu/professional/digital-programs/courses/IoT/phone/index.html
    2. https://swayam.gov.in/nd1_noc19_cs65/preview
    3. https://www.edureka.co/blog/iot-tutorial/
    4. http://www.steves-internet-guide.com/internet-of-things/

- **Software:**
    1. Contiki OS
    2. Node-Red
    3. Proteus
    4. Thinker Cad

CE478: NATURAL LANGUAGE PROCESSING (PE-III)

Description:

This course Natural Language Processing is offered from SWAYAM as noc23_cs45 - Natural Language Processing.

URL: https://onlinecourses.nptel.ac.in/noc23_cs45/preview

Credit and Week:

Teaching Scheme	Week	Theory	Practical	Total
Credit	12	3	1	4
Marks		100	50	150

* Practical component is offered from CHARUSAT

About this course:

This course starts with the basics of text processing including basic pre-processing, spelling correction, language modelling, Part-of-Speech tagging, Constituency and Dependency Parsing, Lexical Semantics, distributional Semantics and topic models. Finally, the course also covers some of the most interesting applications of text mining such as entity linking, relation extraction, text summarization, text classification, sentiment analysis and opinion mining.

Industry Support:

IBM Corporation, Google, JUST AI LIMITED, linguamatics, Amazon

Course Layout:

Week 1: Introduction and Basic Text Processing

Week 2: Spelling Correction, Language Modelling

Week 3: Advanced smoothing for language modelling, POS tagging

Week 4: Models for Sequential tagging – MaxEnt, CRF

Week 5: Syntax – Constituency Parsing

Week 6: Dependency Parsing

Week 7: Distributional Semantics

Week 8: Lexical Semantics

Week 9: Topic Models

Week 10: Entity Linking, Information Extraction

Week 11: Text Summarization, Text Classification

Week 12: Sentiment Analysis and Opinion Mining

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Understand basic text processing, language modelling and POS tagging
CO2	Analyse the various models for sequential tagging
CO3	Understand Syntax Parsing and Dependency Parsing
CO4	Evaluate and Analyse different semantics and topic models
CO5	Create applications demonstrating text classification, summarization and sentiment analysis or opinion mining

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	1	-	2	-	-	-	-	-	-	-	2	1
CO2	2	3	-	1	2	-	-	-	-	-	-	-	2	1
CO3	2	2	1	-	1	-	-	-	-	-	-	-	1	-
CO4	2	2	1	1	-	-	-	-	-	-	-	-	1	-
CO5	3	2	1	2	2	-	-	-	-	-	-	-	3	1

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Books and references:

1. Dan Jurafsky and James Martin. Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics and Speech Recognition. Prentice Hall, Second Edition, 2009. Some draft chapters of the third edition are available online: https://web.stanford.edu/~jurafsky/slp3/

2. Chris Manning and Hinrich Schütze. Foundations of Statistical Natural Language Processing. MIT Press, Cambridge, MA: May 1999.

3. Natural Language Processing with Python, written by Steven Bird, Ewan Klein and Edward Loper.

4. Text Mining with R by Julia Silge and David Robinson

5. Foundations of Statistical Natural Language Processing, Written by Christopher Manning and Hinrich Schütze.

6. Neural Network Methods in Natural Language Processing (Synthesis Lectures on Human Language Technologies, 37) by Yoav Goldberg (Author), Graeme Hirst (Editor)

7. The Oxford Handbook of Computational Linguistics (Oxford Handbooks) 1st Edition by Ruslan Mitkov (Editor)

❖ **Web material:**

1. **https://www.deeplearning.ai/resources/natural-language-processing/**
2. **https://realpython.com/natural-language-processing-spacy-python/**
3. **https://www.ibm.com/topics/natural-language-processing**
4. **https://www.javatpoint.com/nlp**

❖ **Software:**

1. Orange, R and Python
2. NLTK
3. Stanford CoreNLP
4. spaCy
5. Gensim

CRITERIA TO GET A CERTIFICATE

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course.
- Exam score = 75% of the proctored certification exam score out of 100
- Final score = Average assignment score + Exam score

- **YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75. If one of the 2**

criteria is not met, you will not get the certificate even if the Final score >= 40/100.

- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Kharagpur. It will be e-verifiable at nptel.ac.in/noc.

- Only the e-certificate will be made available. Hard copies will not be dispatched.

CE479: ETHICAL HACKING (PE-III)

Description:
This course Ethical Hacking is offered from SWAYAM as noc23_cs75 -Ethical Hacking.

URL: https://onlinecourses.nptel.ac.in/noc23_cs75/preview

Credit and Week:

Teaching Scheme	Week	Theory	Practical	Total
Credit	12	3	1	4
Marks		100	50	150

About this course:

Ethical hacking is a subject that has become very important in present-day context, and can help individuals and organizations to adopt safe practices and usage of their IT infrastructure. Starting from the basic topics like networking, network security and cryptography, the course will cover various attacks and vulnerabilities and ways to secure them. There will be hands-on demonstrations that will be helpful to the participants. The participants are encouraged to try and replicate the demonstration experiments that will be discussed as part of the course.

Industry Support:

TCS, Wipro, CTS, Google, Microsoft, Qualcomm

Course Layout:

Week 1 : Introduction to ethical hacking. Fundamentals of computer networking. TCP/IP protocol stack
Week 2 : IP addressing and routing. TCP and UDP. IP subnets.
Week 3 : Routing protocols. IP version 6.
Week 4 : Installation of attacker and victim system. Information gathering using advanced google search, archive.org, netcraft, whois, host, dig, dnsenum and NMAP tool.
Week 5 : Vulnerability scanning using NMAP and Nessus. Creating a secure hacking environment. System Hacking: password cracking, privilege escalation, application execution. Malware and Virus. ARP spoofing and MAC attack.
Week 6 : Introduction to cryptography, private-key encryption, public-key encryption.
Week 7 : Cryptographic hash functions, digital signature and certificate, applications.
Week 8 : Steganography, biometric authentication, network-based attacks, DNS and Email security.
Week 9 : Packet sniffing using wireshark and burpsuite, password attack using burp suite. Social engineering attacks and Denial of service attacks.

Week 10 : Elements of hardware security: side-channel attacks, physical inclinable functions, hardware trojans.

Week 11 : Different types of attacks using Metasploit framework: password cracking, privilege escalation, remote code execution, etc.Attack on web servers: password attack, SQL injection, cross site scripting.

Week 12 : Case studies: various attacks scenarios and their remedies.

At the end of the course, the students will be able to

CO1	Understand the fundamentals of ethical hacking and its significance in cybersecurity & Describe the basics of computer networking, including TCP/IP protocol stack, IP addressing, and routing.
CO2	Conduct vulnerability scanning using NMAP and Nessus and implement system hacking techniques.
CO3	Understand cryptographic principles, encryption methods, and secure communication techniques.
CO4	Explore advanced security concepts, including steganography, biometric authentication, and network-based attacks.
CO5	Analyze network traffic, conduct ethical hacking attacks, and apply ethical hacking skills in real-world scenarios.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	2	-	1	-	-	1	-	2	1	1	-	-	1	1
CO2	1	1	-	1	3	-	-	-	-	-	2	-	2	-
CO3	1	2	1	-	3	-	-	-	-	-	1	1	2	-
CO4	-	1	1	1	1	-	-	-	-	-	-	-	-	-
CO5	-	1	1	2	1	1	2	1	1	1	-	-	1	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Books and references:

1. Data and Computer Communications -- W. Stallings.
2. Data Communication and Networking -- B. A. Forouzan
3. TCP/IP Protocol Suite -- B. A. Forouzan
4. UNIX Network Programming -- W. R. Stallings

5. Introduction to Computer Networks and Cybersecurity -- C-H. Wu and J. D. Irwin
6. Cryptography and Network Security: Principles and Practice -- W. Stallings

CRITERIA TO GET A CERTIFICATE

- Average assignment score = 25% of average of best 8 assignments out of the total 12 assignments given in the course .

- Exam score = 75% of the proctored certification exam score out of 100 Final score

  Average assignment score + Exam score

- **YOU WILL BE ELIGIBLE FOR A CERTIFICATE ONLY IF AVERAGE ASSIGNMENT SCORE >=10/25 AND EXAM SCORE >= 30/75. If one of the 2 criteria is not met, you will not get the certificate even if the Final score >= 40/100.**

- Certificate will have your name, photograph and the score in the final exam with the breakup. It will have the logos of NPTEL and IIT Kharagpur. It will be e-verifiable at nptel.ac.in/noc.

- Only the e-certificate will be made available. Hard copies will not be dispatched.

CE450: SOFTWARE GROUP PROJECT-V

Credits and Hours:

Teaching Scheme	Theory	Practical	Tutorial	Total	Credit
Hours/week	-	4	-	4	**2**
Marks	-	100	-	100	

Pre-requisite courses:

- Programming Knowledge

Outline of the Course:

- Student at the beginning of a semester may be advised by his/her supervisor (s) for recommended courses.
- Students will work together in a team (at most three) with any programming language.
- Students are required to get approval of project definition from the department.
- After approval of project definition students are required to report their project work on weekly basis to the respective internal guide.
- Project will be evaluated at least once per week in laboratory Hours during the semester and final submission will be taken at the end of the semester as a part of continuous evaluation.
- Project work should include whole SDLC of development of software / hardware system as a solution of particular problem by applying principles of Software Engineering.
- Students have to submit project with following listed documents at the time of final submission.
  a. Project Synopsis
  b. Software Requirement Specification
  c. SPMP
  d. Final Project Report
  e. Project Setup file with Source code
  f. Project Presentation (PPT)
- A student has to produce some useful outcome by conducting experiments or project work.

Total hours (Theory) : 00

Total hours (Lab) : 30

Total hours : 30

Course Outcome (COs):

At the end of the course, the students will be able to:

CO1	Identify and define the computing requirements of a problem to propose its appropriate solution.
CO2	Correlate knowledge of different subjects and apply it to implement solution of the problem.
CO3	Apply engineering and management principles to achieve project goal.
CO4	Prepare technical report and deliver presentation by applying different visualization tools and evaluation metrics.
CO5	Able to communicate effectively with a range of audiences.
CO6	Recognition of the need for and an ability to engage in continuing professional development.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	-	3	-	2	-	-	-	-	2	-	-	-	3	-
CO2	3	-	1	1	3	1	1	-	-	-	-	2	1	1
CO3	-	-	3	-	3	1	1	-	2	-	3	2	3	-
CO4	-	-	-	-	3	-	2	-	2	3	-	-	-	-
CO5	-	-	-	-	-	-	-	-	-	3	-	-	-	-
CO6	-	-	-	-	-	-	3	-	-	-	-	2	-	2

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- **Reference book:**
  1. Books, Magazines & Journals of related topics
- **Web material:**
  1. www.ieeexplore.ieee.org
  2. www.sciencedirect.com
  3. www.elsevier.com
  4. http://spie.org/x576.xml
- **Software:**
  1. ASP.NET
  2. PYTHON/MATLAB
  3. PHP
  4. ANDROID/IOS

CE446: SUMMER INTERNSHIP-II

Credits and Hours:

Teaching Scheme	Project	Practical	Tutorial	Total	Credit
Hours	90	**-**	-	90	**3**
Marks	**150**	-	-	**150**	

Objectives of the Course:

- To get familiar with modern tools and technologies use in company/industry/organization
- To get involved in design, development and testing practices followed in the company/industry/organization
- To enhance their soft-skills, presentation skills, interpersonal skills, documentation skills and office etiquettes required to sustain in company/industry/organization environment
- To participate in teamwork and preferably as part of a multi-disciplinary team
- To make them aware about company/industry/organization best practices, processes and regulations.
- To make them more productive, consistent and punctual.

Outline of the Course:

1.	**Instructional Method and Pedagogy**
	• Summer internship shall be at least 90 hours during the summer vacation only. • Department/Institute will help students to find an appropriate company/industry/organization for the summer internship. • The student must fill up and get approved a Summer Internship Acceptance form by the company and provide it to the Coordinator of the department within the specified deadline. • Students shall commence the internship after the approval of the department Coordinator. Summer internships in research centers is also allowed. • During the entire period of internship, the student shall obey the rules and regulations of the company/industry/organization and those of the University. • Due to inevitable reasons, if the student will not able to attend the internship for few days with the permission of the supervisor, the department Coordinator should be informed via e-mail and these days should be compensated later.

	• The student shall submit two documents to the Coordinator for the evaluation of the summer internship:
	  ◦ Summer Internship Report
	  ◦ Summer Internship Assessment Form
	• Upon the completion of summer internship, a hard copy of "Summer Internship Report" must be submitted through the presentation to the Coordinator by the first day of the new term.
	• The report must outline the experience and observations gained through practical internship, in accordance with the required content and the format described in this guideline. Each report will be evaluated by a faculty member of the department on a satisfactory/unsatisfactory basis at the beginning of the semester.
	• If the evaluation of the report is unsatisfactory, it shall be returned to the student for revision and/or rewriting. If the revised report is still unsatisfactory the student shall be requested to repeat the summer internship.

2.	**Format of Summer Internship Report**
	The report shall comply with the summer internship program principles. Main headings are to be centered and written in capital boldface letters. Sub-titles shall be written in small letters and boldface. The typeface shall be Times New Roman font with 12pt. All the margins shall be 2.5cm. The report shall be submitted in printed form and filed. An electronic copy of the report shall be recorded in a CD and enclosed in the report. Each report shall be bound in a simple wire vinyl file and contain the following sections: • Cover Page • Page of Approval and Grading • Abstract page: An abstract gives the essence of the report (usually less than one page). Abstract is written after the report is completed. It must contain the purpose and scope of internship, the actual work done in the plant, and conclusions arrived at. • TABLE OF CONTENTS (with the corresponding page numbers) • LIST OF FIGURES AND TABLES (with the corresponding page numbers)

- DESCRIPTION OF THE COMPANY/INDUSTRY/ORGANISATION: Summarize the work type, administrative structure, number of employees (how many engineers, under which division, etc.), etc. Provide information regarding
    - Location and spread of the company
    - Number of employees, engineers, technicians, administrators in the company
    - Divisions of the company
    - Your group and division
    - Administrative tree (if available)
    - Main functions of the company
    - Customer profile and market share
- INTRODUCTION: In this section, give the purpose of the summer internship, reasons for choosing the location and company, and general information regarding the nature of work you carried out.
- PROBLEM STATEMENT: What is the problem you are solving, and what are the reasons and causes of this problem.
- SOLUTION: In this section, describe what you did and what you observed during the summer internship. It is very important that majority of what you write should be based on what you did and observed that truly belongs to the company/industry/organization.
- CONCLUSIONS: In the last section, summarize the summer internship activities. Present your observations, contributions and intellectual benefits. If this is your second summer internship, compare the first and second summer internships and your preferences.
- REFERENCES: List any source you have used in the document including books, articles and web sites in a consistent format.
- APPENDICES: If you have supplementary material (not appropriate for the main body of the report), you can place them here. These could be schematics, algorithms, drawings, etc. If the document is a datasheet and it can be easily accessed from the internet, then you can refer to it with the appropriate internet link and document number. In this manner you don’t have to print it

	and waste tons of paper.	
	Total hours (Project):	90
	Total hours:	90

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Ability to integrate existing and new technical knowledge for industrial application.
CO2	Executing work with team and teammates from other disciplines
CO3	Get practices and experience related to professional and ethical issues in the work environment
CO4	Experience of demonstrating the impact of the internship on their learning and professional development.
CO5	Understanding of lifelong learning processes through critical reflection of internship experiences.

Course Articulation Matrix:

	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PO12	PSO1	PSO2
CO1	3	1	-	-	1	-	-	-	-	-	-	-	-	-
CO2	-	-	-	-	-	-	-	-	3	2	1	-	-	2
CO3	-	-	-	-	-	-	1	3	-	1	-	-	-	1
CO4	-	-	-	-	-		3	1	-	-	-	-	1	-
CO5	-		-	-	-	1	-	-	-	-	-	3	-	-

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put “-”

Recommended Study Material:

- ❖ **Reference book:**
    1. Books, Magazines & Journals of related topics

❖ **Web material:**

1. www.ieeexplore.ieee.org
2. www.sciencedirect.com
3. www.elsevier.com
4. http://spie.org/x576.xml

❖ **Software:**

1. ASP.NET
2. PYTHON/MATLAB
3. PHP
4. ANDROID/IOS
5. FLUTTER
6. NODE/REACT NATIVE

B. Tech. (Computer Engineering) Programme

SYLLABI
(Semester – 8)

CHAROTAR UNIVERSITY OF SCIENCE AND TECHNOLOGY

CE451: SOFTWARE PROJECT MAJOR

Credits and Hours:

Teaching Scheme	Theory	Practical	Total	Credit
Hours/week	-	36	36	**18**
Marks	-	600(250+350)	600	

Pre-requisite courses:

- Software Engineering and other relevant courses, Development tools and languages, soft skill.

Outline of the Course:

- Students work with industry/organization4-to-6 months for development or research project.
- The definitions are verified and guidance are given to take the project at next level and can help industry, society or environment.
- The external guide (at industry/Organization) and internal guide (Institute faculty) continuously monitor the students' work and project file is maintained per group to document all the measurable work.
- Project work should include whole SDLC/Agile of development of software / hardware system as solution of particular problem by applying principles of Software Engineering.
- Project is evaluated twice during the semester by internal faculty of the university as a part of continuous evaluation.
- Final evaluation at the end of the semester is done by Industry experts or faculties from reputed university.
- Students have to submit SRS, SPMP, Design documents, Code and Test Cases in form of Project report.
- The feedback of the students by external guide and Evaluator are taken during the semester to improve the teaching learning and evaluation process.

Course Outcome (COs):

At the end of the course, the students will be able to

CO1	Identify and justify/analyse the requirements of the projects with enhancement of the required tools and technology by individual and team.
CO2	Solve challenging projects for commercial, societal and environment benefit.
CO3	Apply the engineering knowledge to full fill the requirements of the projects pertaining to any discipline.
CO4	Explain the importance of planning, documentation, punctuality and work ethics.
CO5	Document the work which is carried out in proper format with industry standards.
CO6	Showcase the soft skill.

Course Articulation Matrix:

	PO01	PO02	PO03	PO04	PO05	PO06	PO07	PO08	PO09	PO10	PO11	PO12	PSO1	PSO2
CO1	3	3	-	2	3	-	1	-	3	1	1	1	3	1
CO2	2	-	3	-	-	3	2	-	-	-	-	-	2	2
CO3	3	2	1	1	1	-	-	-	1	-	2	1	3	1
CO4	3	1	2	-	-	-	-	3	-	-	1	2	2	3
CO5	3	-	2	2	-	-	-	-	-	2	-	3	1	1
CO6	1	-	1	1	-	-	-	3	1	3	1	1	1	3

Enter correlation levels 1, 2 or 3 as defined below:

1: Slight (Low) 2: Moderate (Medium) 3: Substantial (High)

If there is no correlation, put "-"

Recommended Study Material (Suggested by Internal or External guide):

- Reading Materials, web materials, Project reports with full citations.
- Books, magazines & Journals of related topics.
- Various software tools and programming languages compiler related to topic.
- ❖ **Web Materials**:
  1. www.ieeexplore.ieee.org
  2. www.sciencedirect.com
  3. www.elsevier.com